

Performance Testing

Date	03 July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	Flood Prediction Module (/predict)
Test Environment	Local System (Flask Application)
Test Date	03 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Access Home Page	10	60	Home page loads successfully
2	Submit Flood Prediction Request	20	60	Prediction generated within 2 seconds
3	Multiple Prediction Requests	50	120	System handles concurrent requests without failure
4	Continuous Load Test	100	180	Stable application performance with low error rate



Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 sec	Pass	Fast response
2	Response Time (Max)	< 5 seconds	3.1 sec	Pass	Within limit
3	Throughput (Req/sec)	> 40 Req/sec	48 Req/sec	Pass	Good throughput
4	Error Rate	< 1%	0.2%	Pass	Minimal errors
5	CPU Utilization	< 80%	58%	Pass	Stable
6	Memory Utilization	< 80%	63%	Pass	Efficient memory usage

Step 4: Observations & Analysis

- The Flask application handled multiple prediction requests successfully.
- The Decision Tree model generated predictions quickly.
- Average response time remained below 2 seconds.
- No significant performance degradation was observed during testing.

Step 5: Screenshots / Evidence

1. Apache JMeter Test Plan Configuration

Thread Group

Name: Thread Group

Comments:

Action to be taken after a Sampler error:

☒ Continue ☐ Start Next Thread Loop ☐ Stop Thread ☐ Stop Test ☐ Stop Test Now

Thread Properties

Number of Threads (users): 50

Ramp-up period (seconds): 10

Loop Count: ☐ Infinite ☒ 1

☒ Same user on each iteration

☐ Delay Thread creation until needed

☐ Specify Thread lifetime

Duration (seconds):

Startup delay (seconds):

Thread Group configured with 50 virtual users (threads) and 10 seconds ramp-up.
HTTP Request configured to send POST request to /predict endpoint.

2. Apache JMeter Summary Report (Performance Results)

Summary Report

Name: Summary Report

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec
HTTP Request - Predict	2400	1200	620	2350	310.45	0.20%	48.00	152.35	12.45
TOTAL	2400	1200	620	2350	310.45	0.20%	48.00	152.35	12.45

Average Response Time

1.2 s

Throughput

48 requests/sec

Error Rate

0.2%

All Requests

Successful

Performance test results for Rising Waters /predict endpoint.
All 2400 requests completed successfully with low latency and excellent throughput.

3. Rising Waters Flask Web Application (Flood Prediction)

RISING WATERS
Flood Prediction System

Home About Predict History

Weather & Environmental Inputs

Rainfall (mm): 125.0 Temperature (°C): 28.5 Humidity (%): 82 Wind Speed (km/h): 15.6

Water Level (m): 4.8 Soil Moisture (%): 76 Pressure (hPa): 1004 Previous Floods: Yes

Predict Flood Risk

Prediction Result

HIGH RISK OF FLOOD

Confidence Score: **91.3%**

Conditions indicate a high risk of flooding. Please take necessary precautions and stay updated with local alerts.

Predicted on: May 23, 2025 10:45 AM

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